Hello folks!✨

Quick questions before we start

1. How many students here?

2. How many of you here are from Kerala?

3. how many have some background in AI/ML?

4. How many here knows python?

5. How may knows Html/CSS ?

Introduction to

Large Language Models

What comes to your mind when you hear the word

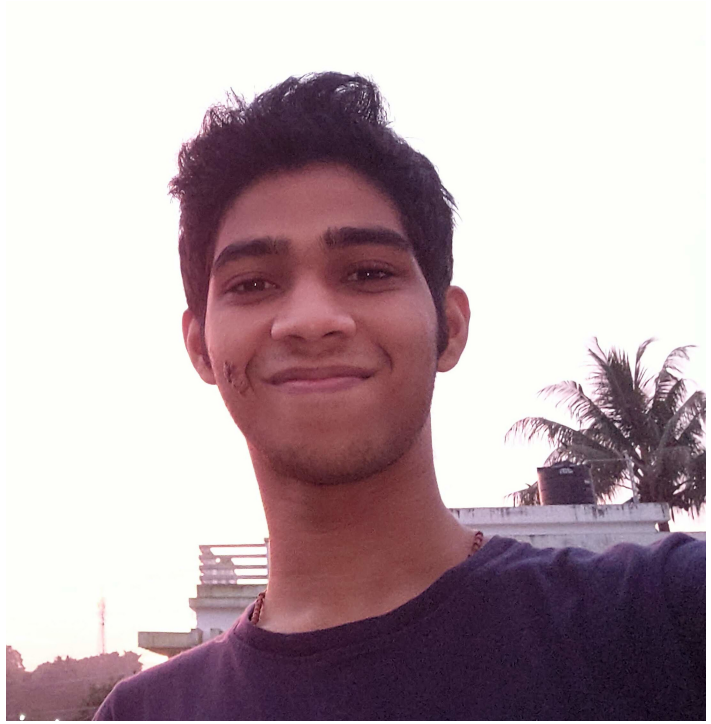
Gen-AI

- OpenAI?
- Devin ?
- SORA?

...

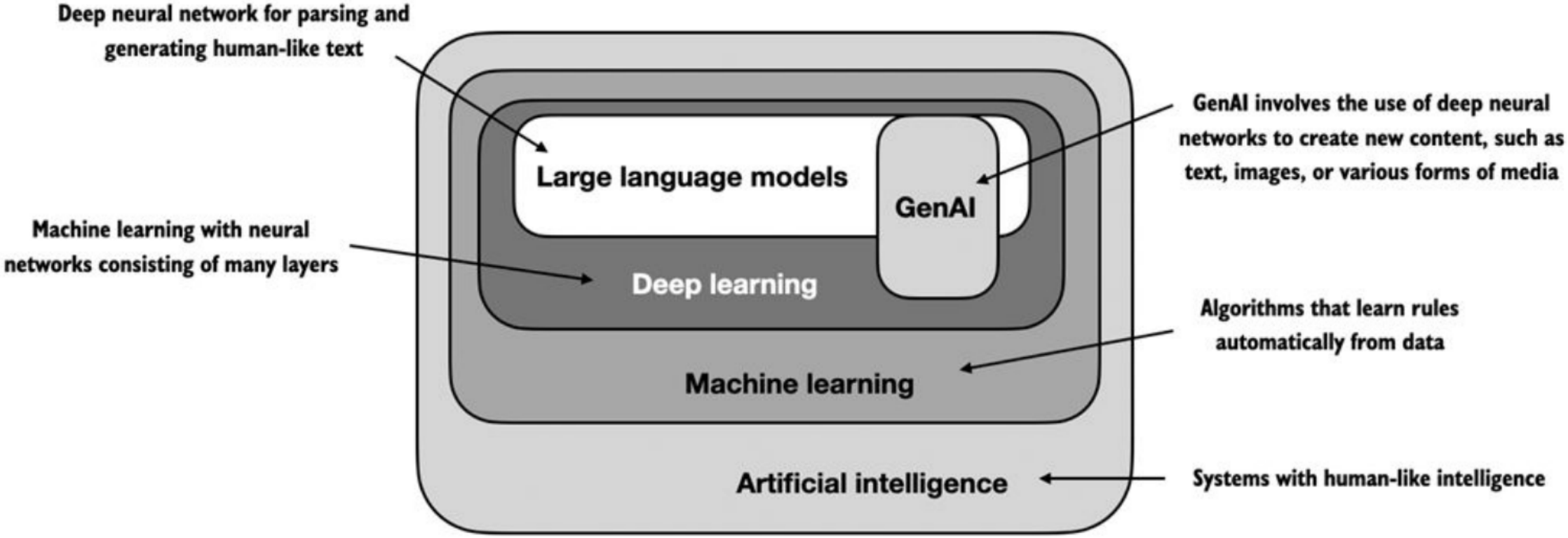
Aldrin Jenson

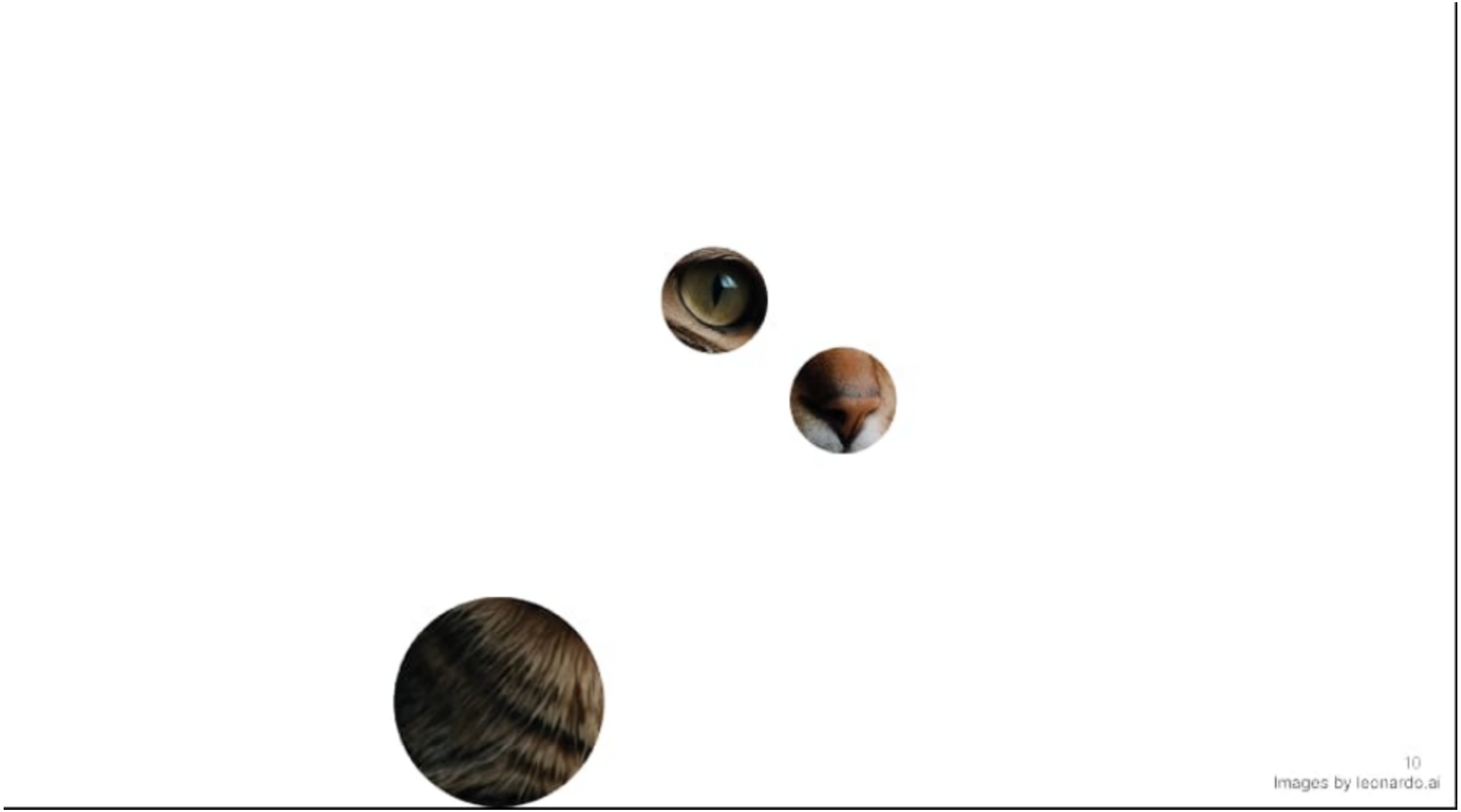
- Software Engineer at **Thoughtworks**
- Focussing on R&D in **Generative AI**
- Volunteer at FOSS United



@aldrinjenson

@aldrinjenson


Deep neural network for parsing and generating human-like text
Large language models
GenAI
GenAI involves the use of deep neural networks to create new content, such as text, images, or various forms of media
Machine learning with neural networks consisting of many layers
Deep learning
Algorithms that learn rules automatically from data
Machine learning
Artificial intelligence
Systems with human-like intelligence



10
Images by leonardo.ai


Discriminative AI
Given features,
AI will be able to
predict a class.
Image by leonardo.ai

Give me a cat dressed up formally


Generative AI
Given a class, you will be able to create image with some noise (creative element)
11
Images by leonardo.ai

What is Generative AI?

- GenAI is a type of Artificial Intelligence that creates new content based on what it has learned from existing content.
- The process of learning from existing content is called training and results in the creation of a statistical model.
- When given a prompt, GenAI uses this statistical model to predict what an expected response might be–and this generates new content.

https://www.youtube.com/watch?v=G2fqAlgmoPo

Classifications

Tools & Applications

Text Generation



Key points

- Generating content
- Translation
- Summarization
- Question Answering
- Grammar correction
- Documentation
- Generating blog content

OpenAI









Image Generation

- Creating realistic images
- Customizing image
- Super resolution
- Text remover
- Remove background,.etc



ClipDrop

DALL·E



Video Generation

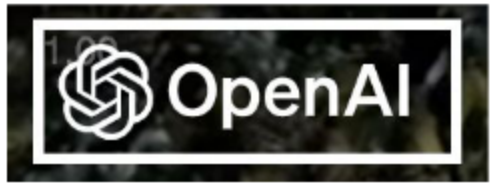
- Create talking avatars with realistic including hand gestures ,emotions
- Animation
- Customize the voice











Audio Generation

- Given the text input,you can generate highly realistic, multilingual speech as well as other audio - including music, background noise and simple sound effects
- Nonverbal communications like laughing, sighing and crying

suno-ai/**bark** 

IIElevenLabs

Code Generation



Key points

- AI pair programmer
- Offers autocomplete-style
- Provide insights into the design and structure of the code
- Identify the areas may need refactoring
- Generate source code documentation











But Why? 🤔

HOME / NEWS / GENERAL NEWS

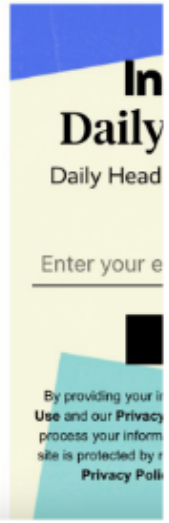
Tyler Perry Put $800 Million Studio Expansion 'Indefinitely on Hold' After Seeing OpenAI Sora's Capabilities

Perry, who recently used AI in two films, fears "a lot of jobs are going to be lost."

BY SAMANTHA BERGESON

FEBRUARY 23, 2024 11:45 AM





https://www.hollywoodreporter.com/business/business-news/tyler-perry-ai-alarm-1235833276/

- Printing Process(~1450)
- Electricity(~1880)
- Assembly Line(~1930)
- Personal Computing(~1930)
- Mobile Computing(~2007)

- Generative AI(~2022)

- Any questions so far?

What are LLMs

- **Very large** AI models designed to understand and generate **human-like text**

- applications include natural language understanding, text generation, translation, summarization etc
- multiple layers of neural networks, typically utilizing transformer architectures.

Few major examples

- ChatGPT
- https://indicsubtitler.in/livetranscribe
- https://www.chatpdf.com/
- https://www.perplexity.ai/
- https://cursor.sh

Any questions.?

Now

Using LLMs to code!

Let's goooo

Prompting

You are one of the best speakers in the world. You create engaging and attractive presentations for talks and webinars. Your job is to assist me in creating a presentation for an event. You will give content, and ideas for images for the presentation.

Topic - 'GPT'
Subtopics - 'What is GPT, How chatGPT uses GPT, Limitations of GPT, Threats of GPT'

Prepare a presentation for an talk on the topic.

Make sure to follow the below rules while generating slides,
- First slide should be agenda.
- The total number of slides should not exceed 4 slides.
- If subtopics are present, use them to frame agenda.

Reply in the following JSON format,

```
{
        "slides": [
        {       "slideNumber": "",
                "slideTitle": "",
                "slideContent": "<Content of the slide>",
                "Imageldeas": [“<Image idea for the slide>”]
        }
]
}
```

Conclusion

Thank You ✨

Shout out to

- Sree Laxmi
- Srilakshmi
- IB Computing
- Titty Jacob Sir(MEC)

@aldrinjenson

aldrinjenson.tech

This presentation was built using Free Software!

Resources

- https://www.gnu.org/gnu/manifesto.html
- https://www.debian.org/
- https://www.fsf.org/
- https://www.youtube.com/watch?v=vwG5N8SCmZ8
- https://indiafoss.net/2023
- https://en.wikipedia.org/wiki/Free_software_movement